# Hospital Assignment Problem using Maximum Flow

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# Problem Statement

Network flow issues come up in dealing with natural disasters and other crises, since major unexpected events often require the movement and evacuation of large numbers of people in a short amount of time.

Consider the following scenario. Due to large-scale flooding in a region, paramedics have identified a set of $n$ injured people distributed across the region who need to be rushed to hospitals. There are $k$ hospitals in the region, and each of the $n$ people needs to be brought to a hospital that is within a half-hour's driving time of their current location (so different people will have different options for hospitals, depending on where they are right now).

At the same time, one doesn't want to overload any one of the hospitals by sending too many patients its way. The paramedics are in touch by cell phone, and they want to collectively work out whether they can choose a hospital for each of the injured people in such a way that the load on the hospitals is balanced: Each hospital receives at most $\lceil n/k \rceil$ people.

Give a polynomial-time algorithm that takes the given information about the people's locations and determines whether this is possible.

# Problem Recap

There are $n$ injured people, each needing to be assigned to one hospital within a 30-minute drive.

There are $k$ hospitals, and no hospital can be assigned more than $\lceil n/k \rceil$ people.

The task: efficiently decide whether it is possible to assign each person to a hospital within their reach such that no hospital is overloaded.

# Algorithm Overview

We model this as a **maximum flow problem** in a bipartite graph. The key insight is to create a flow network where:

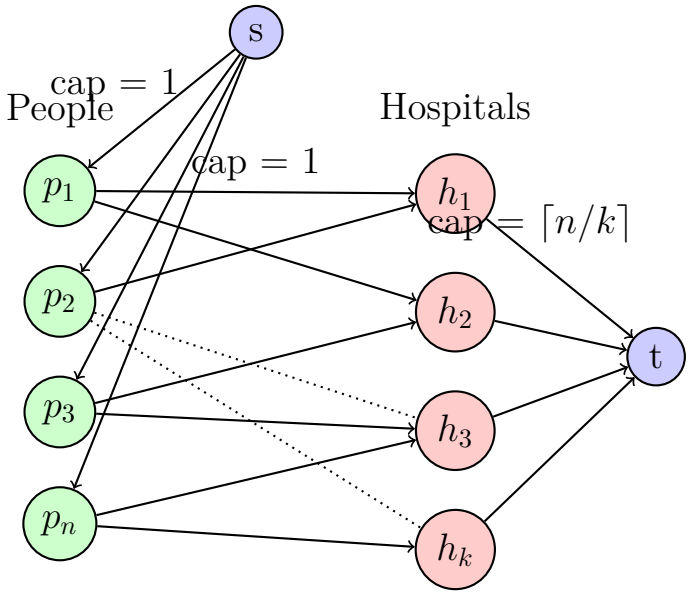
- People and hospitals form two partitions of a bipartite graph
- Edges connect people to hospitals they can reach
- Capacities enforce the constraints

# Flow Network Construction

We construct a directed flow network $G = (V, E)$ with capacities $c : E \to \mathbb{R}^+$ as follows:

1. **Nodes:**
   - $s$: source node
   - $P = \{p_1, p_2, \ldots, p_n\}$: people nodes
   - $H = \{h_1, h_2, \ldots, h_k\}$: hospital nodes
   - $t$: sink node
2. **Edges and Capacities:**
   - For each person $p_i \in P$: edge $(s, p_i)$ with capacity 1
   - For each hospital $h_j \in H$: edge $(h_j, t)$ with capacity $\lceil n/k \rceil$
   - For each person $p_i$ and hospital $h_j$: if $h_j$ is within 30 minutes of $p_i$, add edge $(p_i, h_j)$ with capacity 1

# Graphical Representation



# Algorithm Construction

**Input:**

- A list of $n$ people. For each person $i$, we have a list $H_i$ of hospitals they can reach.
- The number of hospitals $k$.

**Output:** "YES" if a balanced assignment is possible, "NO" otherwise.

**Steps:**

1. **Create a Flow Graph:**
   - Create a source node $s$ and a sink node $t$.
   - Create a node for each of the $n$ people.
   - Create a node for each of the $k$ hospitals.
   - Calculate the capacity for each hospital: $C = \lceil n/k \rceil$.
2. **Add Edges:**
   - For each person $i \in \{1, \ldots, n\}$, add an edge from $s$ to $person_i$ with capacity 1.
   - For each hospital $j \in \{1, \ldots, k\}$, add an edge from $hospital_j$ to $t$ with capacity $C$.
   - For each person $i$ and for each hospital $j \in H_i$, add an edge from $person_i$ to $hospital_j$ with capacity 1.
3. **Compute Maximum Flow:** Run a standard polynomial-time Max-Flow algorithm (such as Edmonds-Karp) on the graph with source $s$ and sink $t$.
4. **Result:** Let $F$ be the value of the maximum flow obtained.

- If $F = n$, output "YES". A valid balanced assignment exists.
- Otherwise (if $F < n$), output "NO". It is impossible to assign all people without violating constraints.

# Complexity Analysis

- The graph has $O(n + k)$ nodes, which is $O(n)$ since $k \leq n$ in a practical scenario.
- The number of edges is:
  - $n$ edges from the source,
  - $k$ edges to the sink,
  - At most $n \times k$ edges between people and hospitals, which in the worst case is $O(n^2)$.
- Using the Edmonds-Karp algorithm, which runs in $O(VE^2)$ time, the overall worst-case time complexity is

$$O((n + k) \cdot (n \cdot k)^2) = O(n^5)$$

  as $k \leq n$.
- Thus, the algorithm runs in polynomial time, although the complexity is high in the worst case.

# Examples of Different Cases

- **Case 1: Perfect Feasible Assignment (Flow = n)**
  There are 4 injured people and 2 hospitals. Each hospital can take $\lceil 4/2 \rceil = 2$ patients. Reachability: P1 $\rightarrow$ H1, H2; P2 $\rightarrow$ H1; P3 $\rightarrow$ H2; P4 $\rightarrow$ H1, H2. Max flow = 4 = n $\Rightarrow$ feasible.

- **Case 2: Uneven Reachability (Still Feasible)**
  3 injured people, 2 hospitals, capacity $\lceil 3/2 \rceil = 2$. Reachability: P1 $\rightarrow$ H1; P2 $\rightarrow$ H1; P3 $\rightarrow$ H2. Max flow = 3 = n $\Rightarrow$ feasible.

- **Case 3: Infeasible Due to Hospital Overload (Flow $<$ n)**
  4 people, 2 hospitals, capacity 2 each. P1, P2, P3 can only reach H1; P4 $\rightarrow$ H2. Max flow = $3 < 4 \Rightarrow$ not feasible.

- **Case 4: Infeasible Due to Connectivity (Distance Constraint)**
  4 people, 2 hospitals, capacity 2 each. P1 $\rightarrow$ H1; P2 $\rightarrow$ H1; P3 $\rightarrow$ none; P4 $\rightarrow$ H2. Max flow = $3 < 4 \Rightarrow$ not feasible.

- **Case 5: Too Many Hospitals (Trivial Feasibility)**
  4 people, 4 hospitals, capacity 1 each. Each person reaches a unique hospital. Max flow = 4 = n $\Rightarrow$ feasible.

- **Case 6: Overprovisioned Capacity (Always Feasible)**
  4 people, 2 hospitals, capacity 3 each. All can reach both hospitals. Max flow = 4 = n $\Rightarrow$ feasible.

- **Case 7: Edge Case – All Reach Single Hospital**
  3 people, 2 hospitals, capacity 1 each. All only can reach H1. Max flow $= 1 < 3 \Rightarrow$ not feasible.

- **Case 8: Minimum Flow (No Assignments Possible)**
  3 people, 2 hospitals, none reachable due to constraints. Max flow $= 0 < 3 \Rightarrow$ not feasible.

The End

